Algorithms

Lecture #42

Syed Hamed Raza

Floyd-Warshall Algorithm

Floyd-Warshall Algorithm

- The algorithm dates back to the early 60's.
- As with other dynamic programming algorithms, the genius of the algorithm is in the clever recursive formulation of the shortest path problem.
- For a path $p = \langle v_1, v_2, \ldots, v_l$, we say that the vertices $v_2, v_3, \ldots, v_{l-1}$ are the intermediate vertices of this path.

Floyd-Warshall Algorithm

Floyd-Warshall Algorithm

Formulation:

- Define $d_{ij}^{(k)}$ to be the shortest path from i to j such that any intermediate vertices on the path are chosen from the set {1, 2, . . . , k}.
- The path is free to visit any subset of these vertices and in any order.

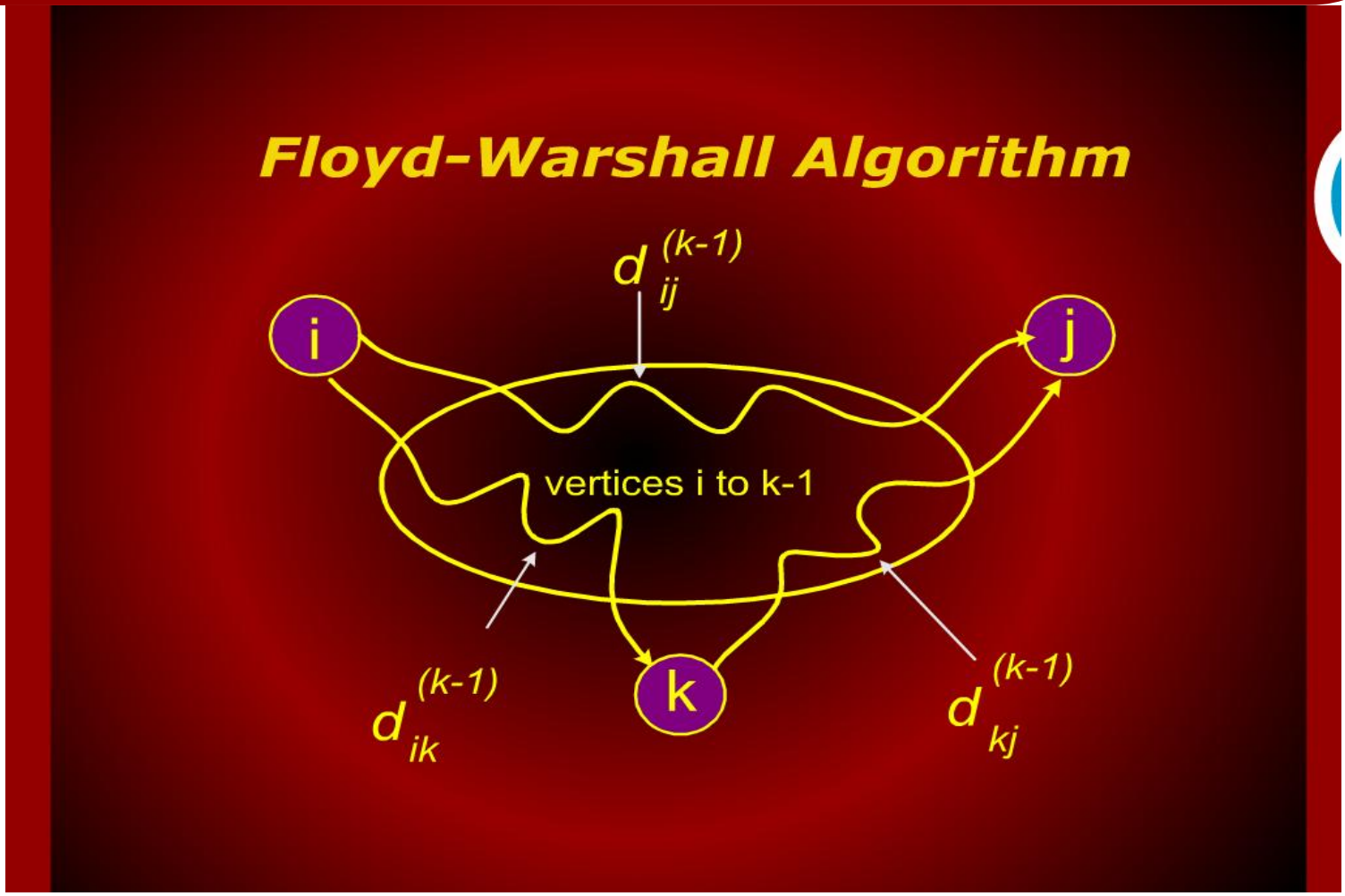
Floyd-Warshall Algorithm

Floyd-Warshall Algorithm

- How do we compute $d_{ij}^{(k)}$ assuming we already have the previous matrix $d^{(k-1)}$?

- There are two basic cases:
  1. Don't go through k at all
  2. Do go through k

Floyd-Warshall Algorithm

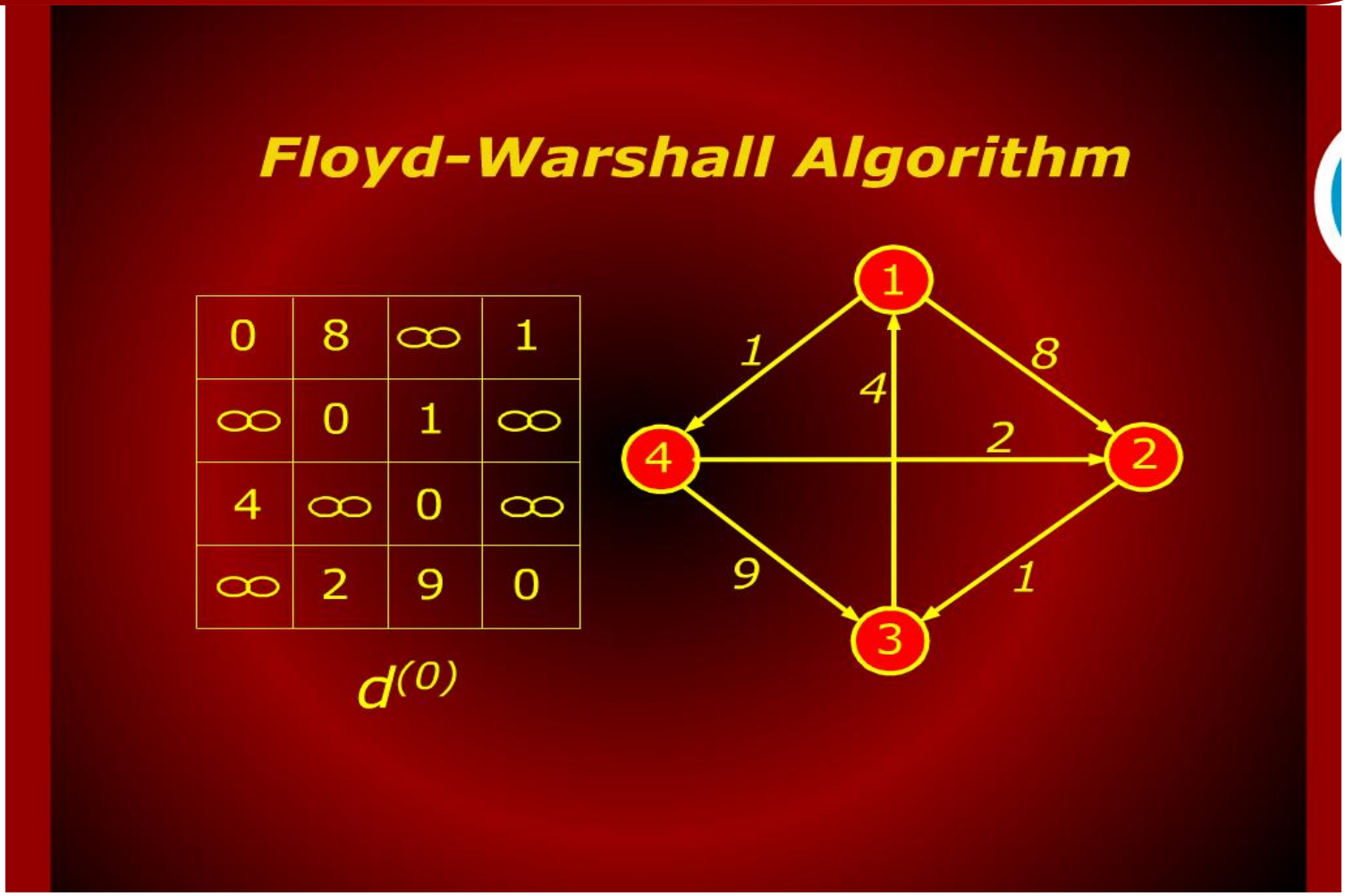


Floyd-Warshall Algorithm

Floyd-Warshall Algorithm

1. Don't go through k at all
   - Then the shortest path from i to j uses only intermediate vertices {1, 2, . . . , k–1} Hence the length of the shortest is $d_{ij}^{(k-1)}$

Floyd-Warshall Algorithm



Floyd-Warshall Algorithm

Floyd-Warshall Algorithm

2. Do go through k

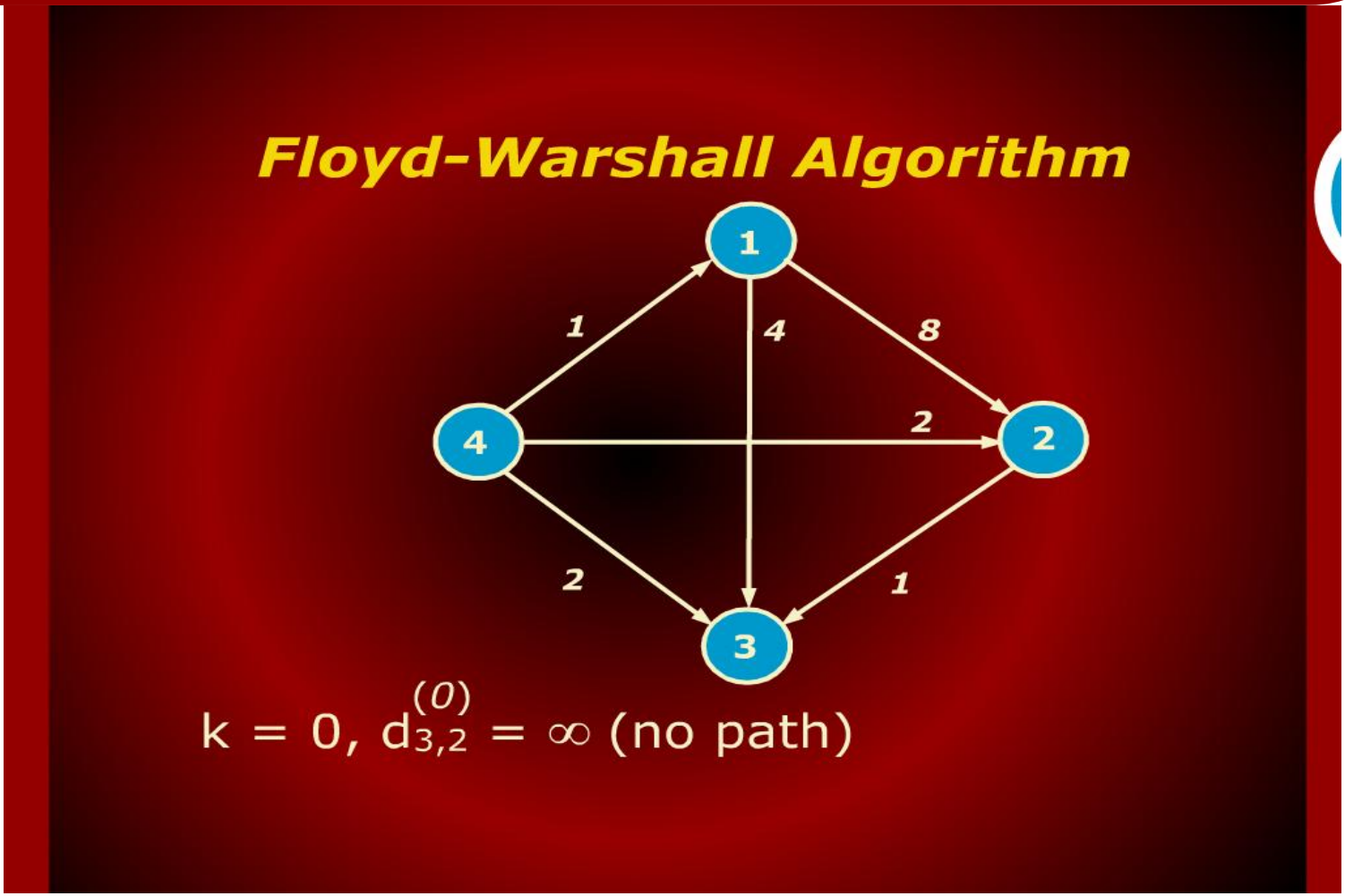
- First observe that a shortest path does not go through the same vertex twice, so we can assume that we pass through k exactly once. That is, we go from i to k and then from k to j.
- In order for the overall path to be as short as possible, we should take the shortest path from i to k and the shortest path from k to j.

Floyd-Warshall Algorithm

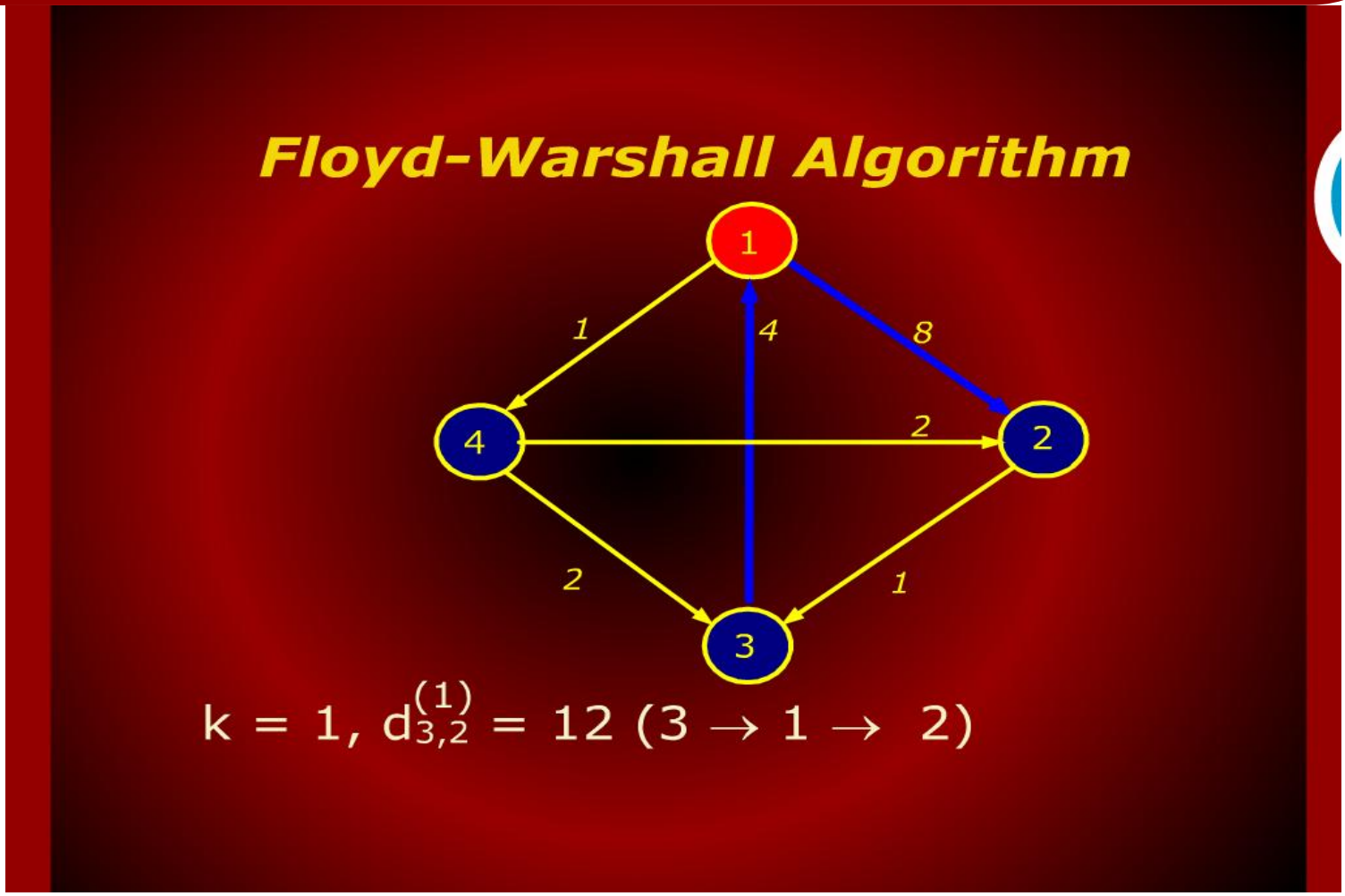
Floyd-Warshall Algorithm

2. Do go through k
   - Since each of these paths uses intermediate vertices $\{1, 2, \ldots, k - 1\}$, the length of the path is $d_{ij}^{(k-1)} + d_{kj}^{(k-1)}$

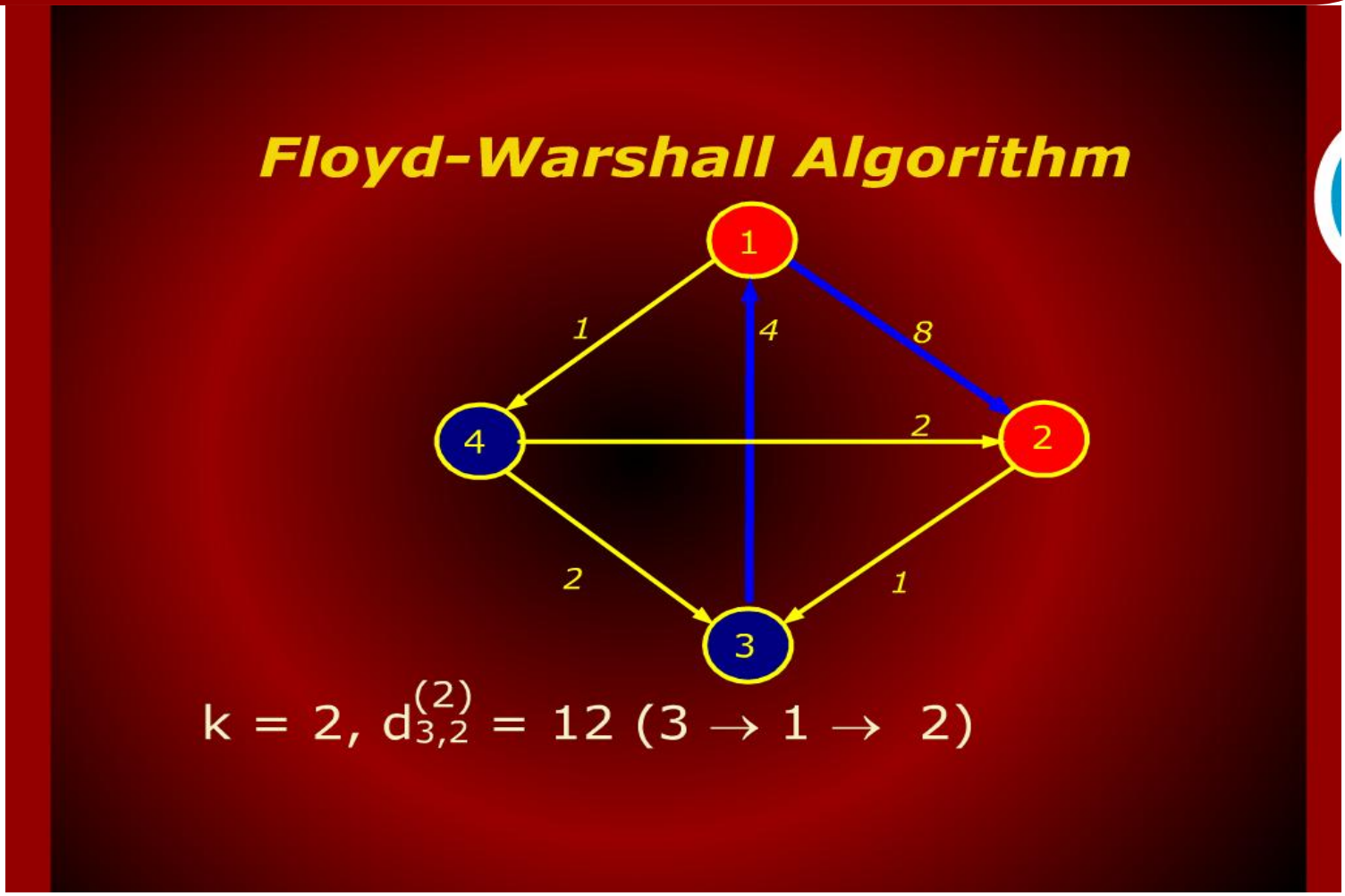
Floyd-Warshall Algorithm



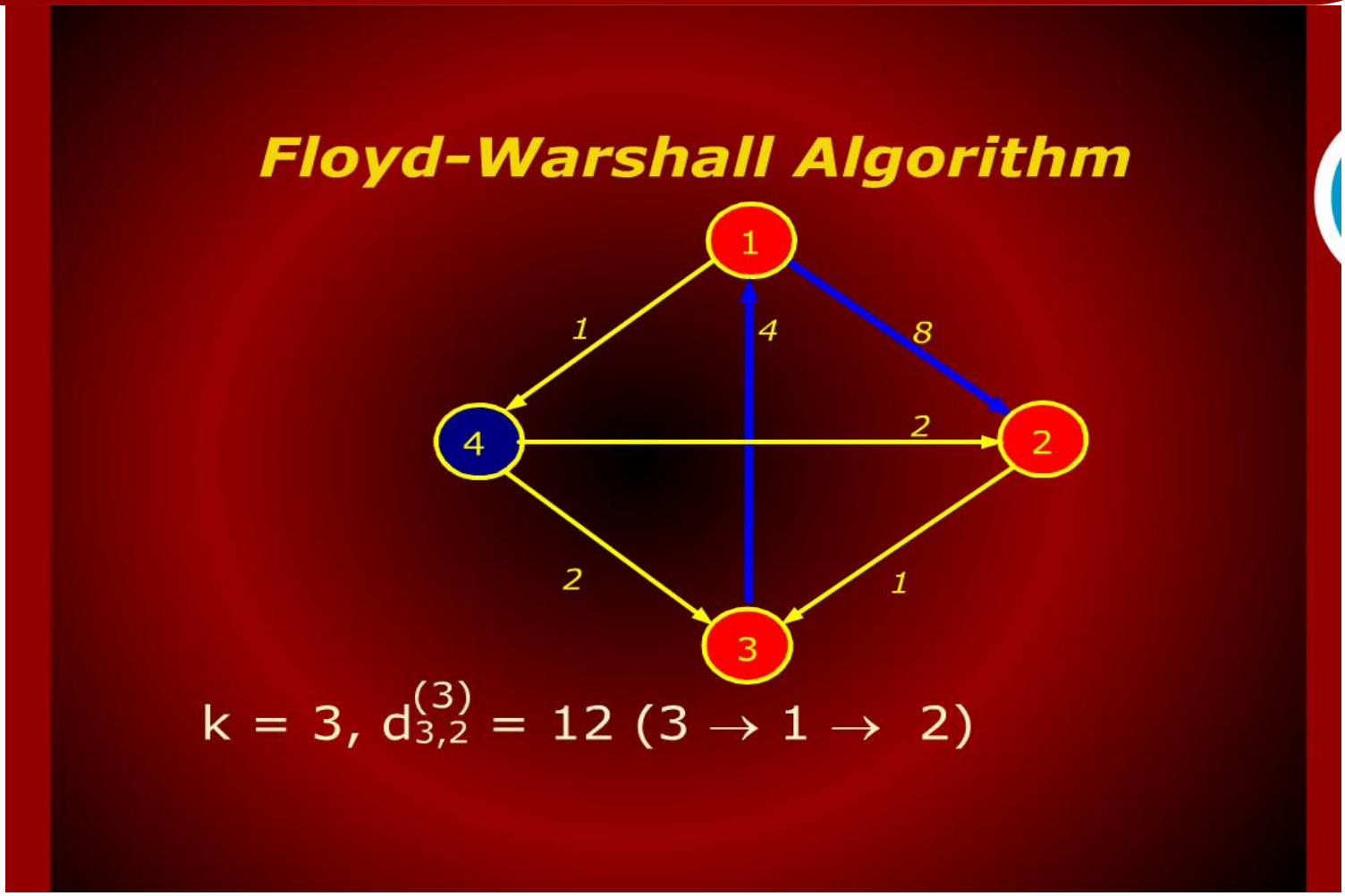
Floyd-Warshall Algorithm



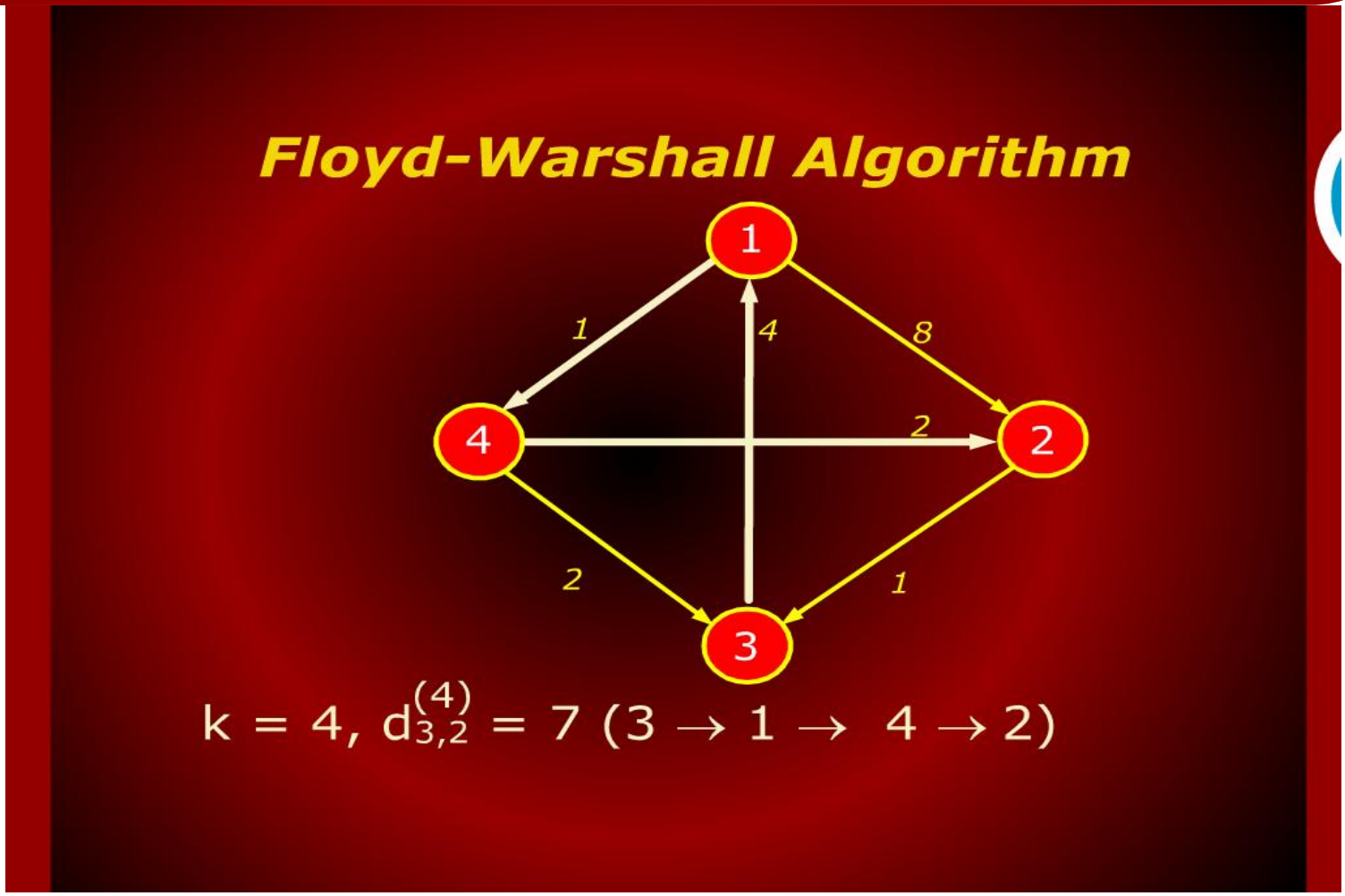
Floyd-Warshall Algorithm



Floyd-Warshall Algorithm



Floyd-Warshall Algorithm



Floyd-Warshall Algorithm

Floyd-Warshall Algorithm

- This suggests the following recursive (DP) formulation:

$$d_{ij}^{(0)} = w_{ij}$$

$$d_{ij}^{(k)} = \min\left(d_{ij}^{(k-1)}, d_{ik}^{(k-1)}, d_{kj}^{(k-1)}\right)$$

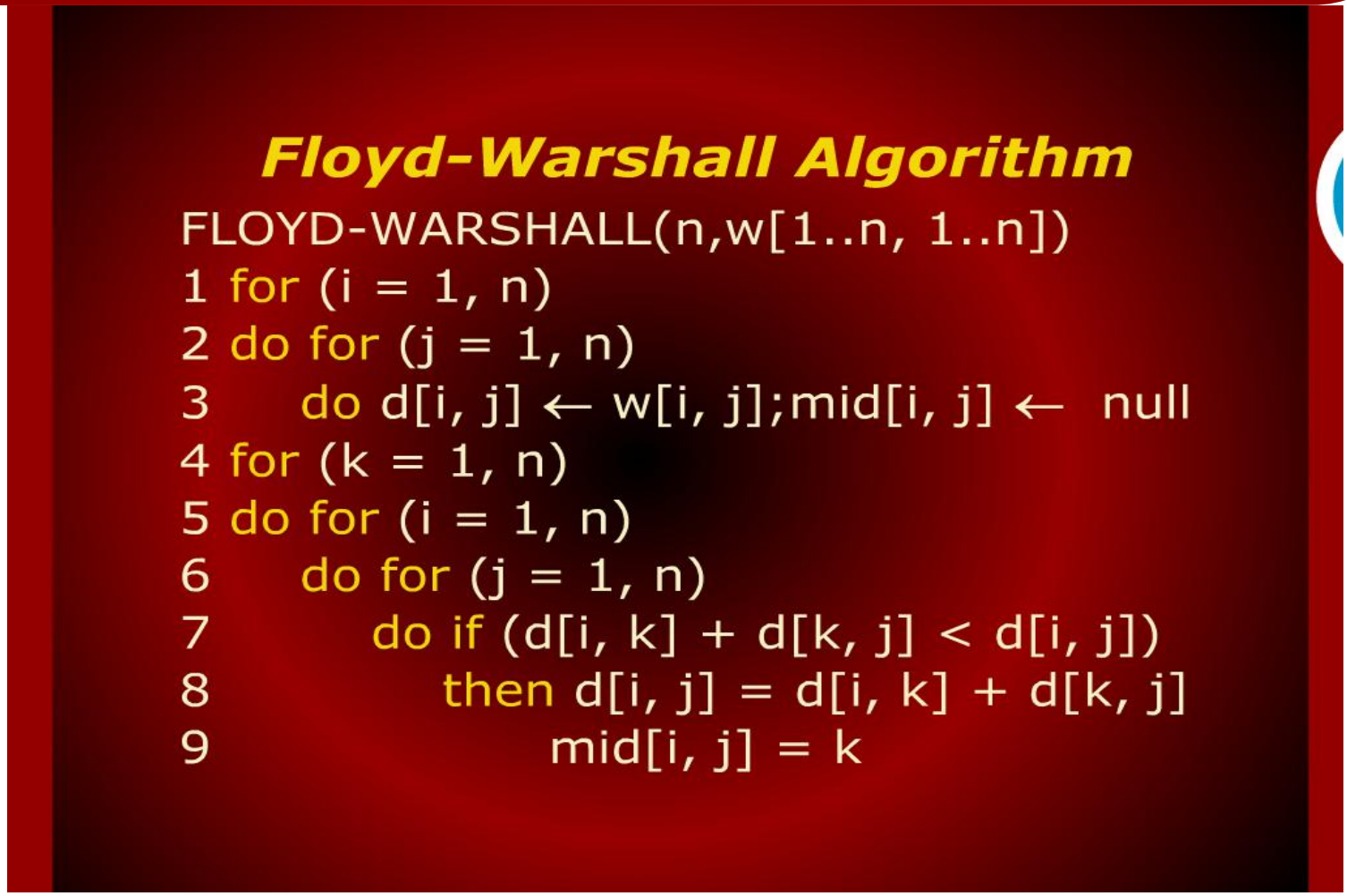
- The final answer is $d_{ij}^{(n)}$ because this allows all possible vertices as intermediate vertices.

Floyd-Warshall Algorithm

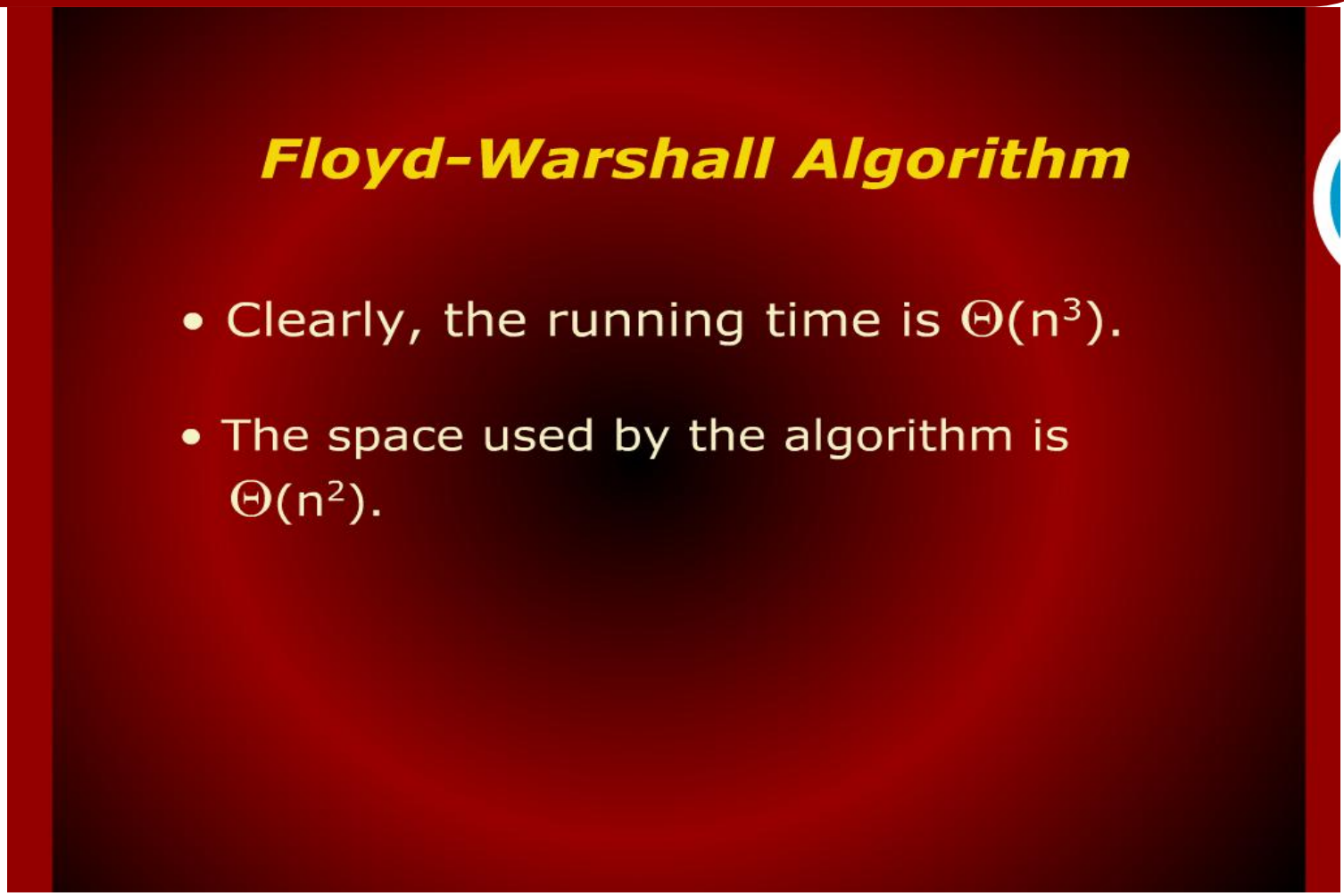
Floyd-Warshall Algorithm

- As is the case with DP algorithms, we will avoid recursive evaluation by generating a table for $d_{ij}^{(k)}$.
- The algorithm also includes mid-vertex pointers stored in mid[i, j] for extracting the final path.

Floyd-Warshall Algorithm



Floyd-Warshall Algorithm



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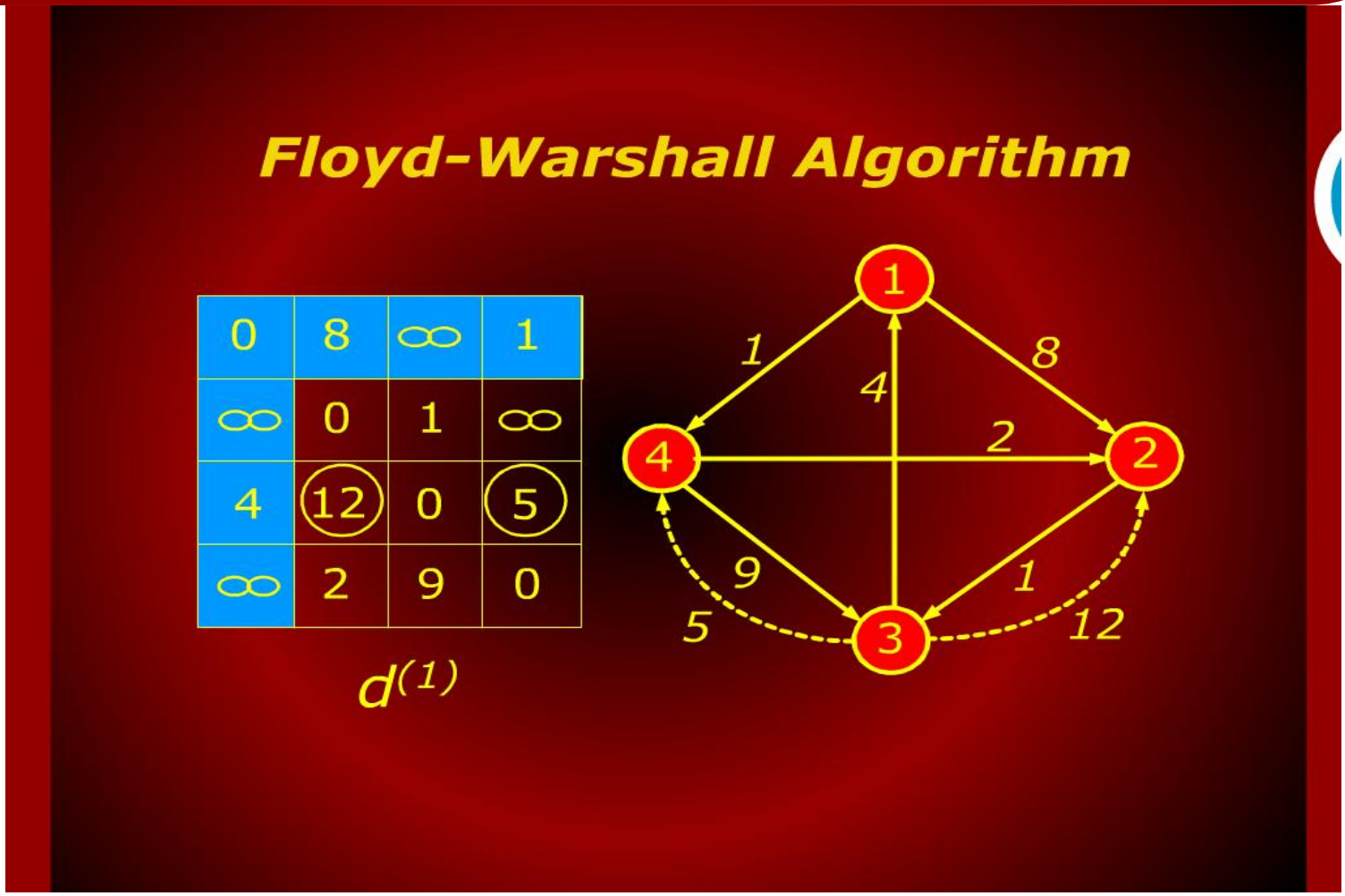

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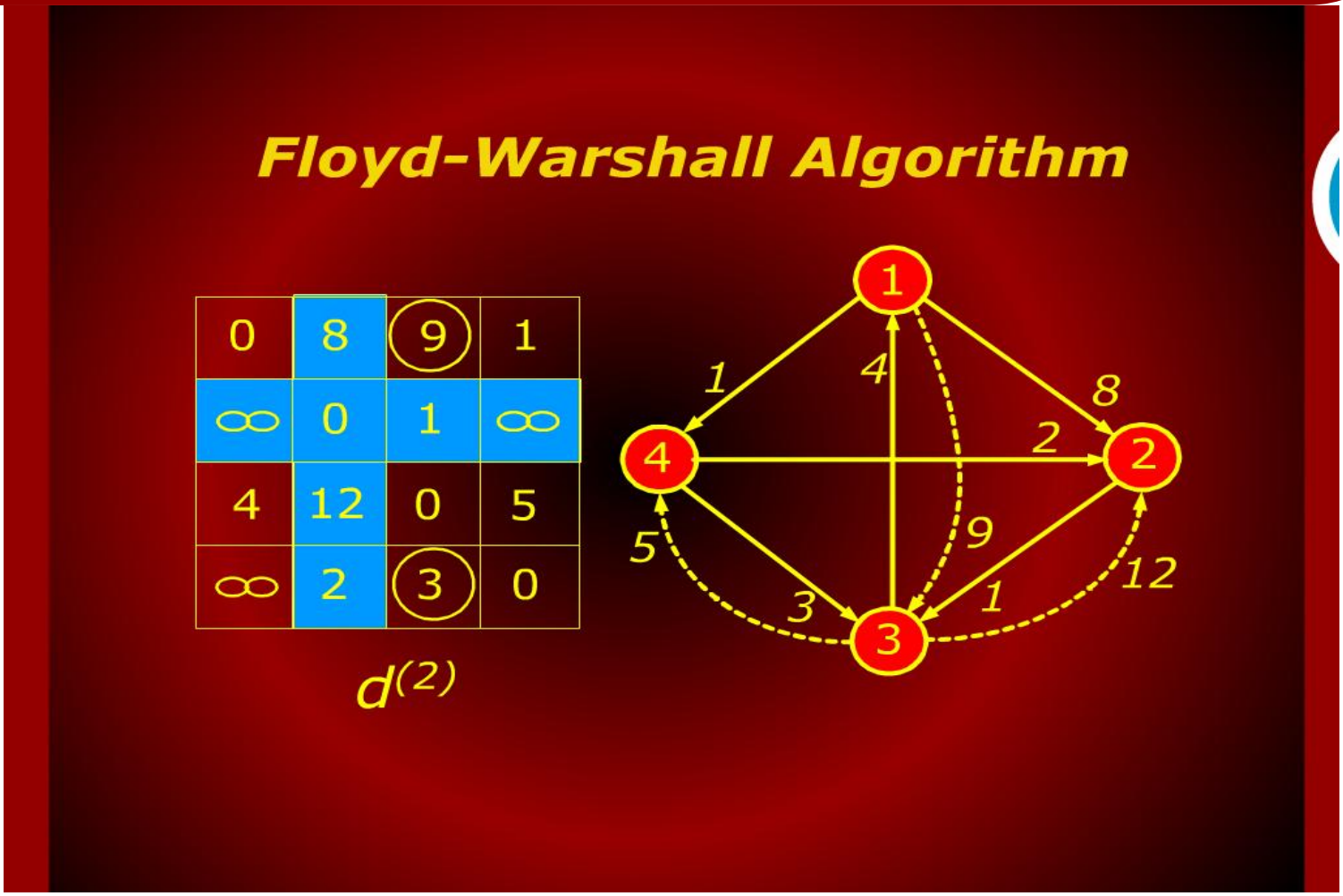

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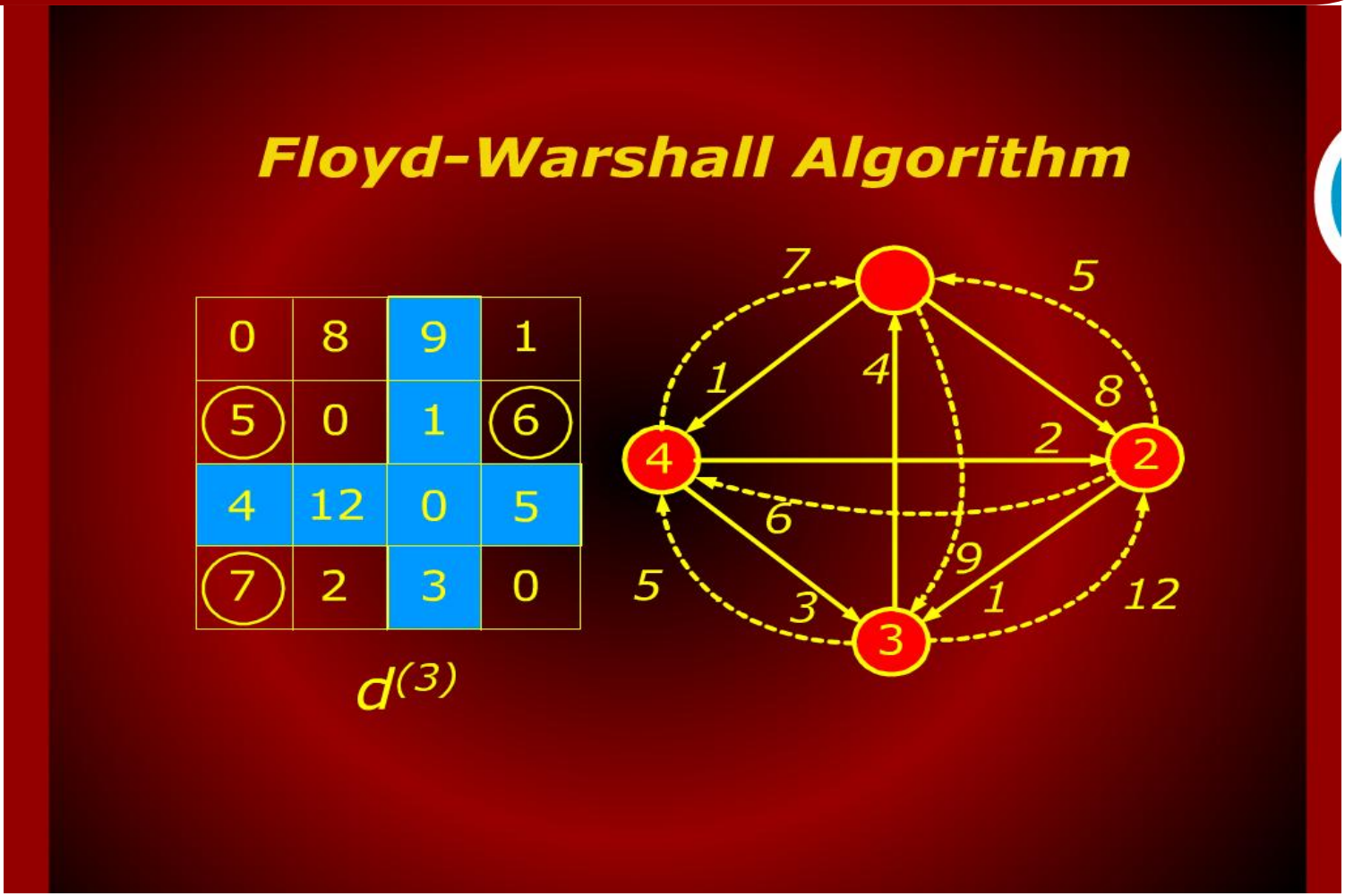

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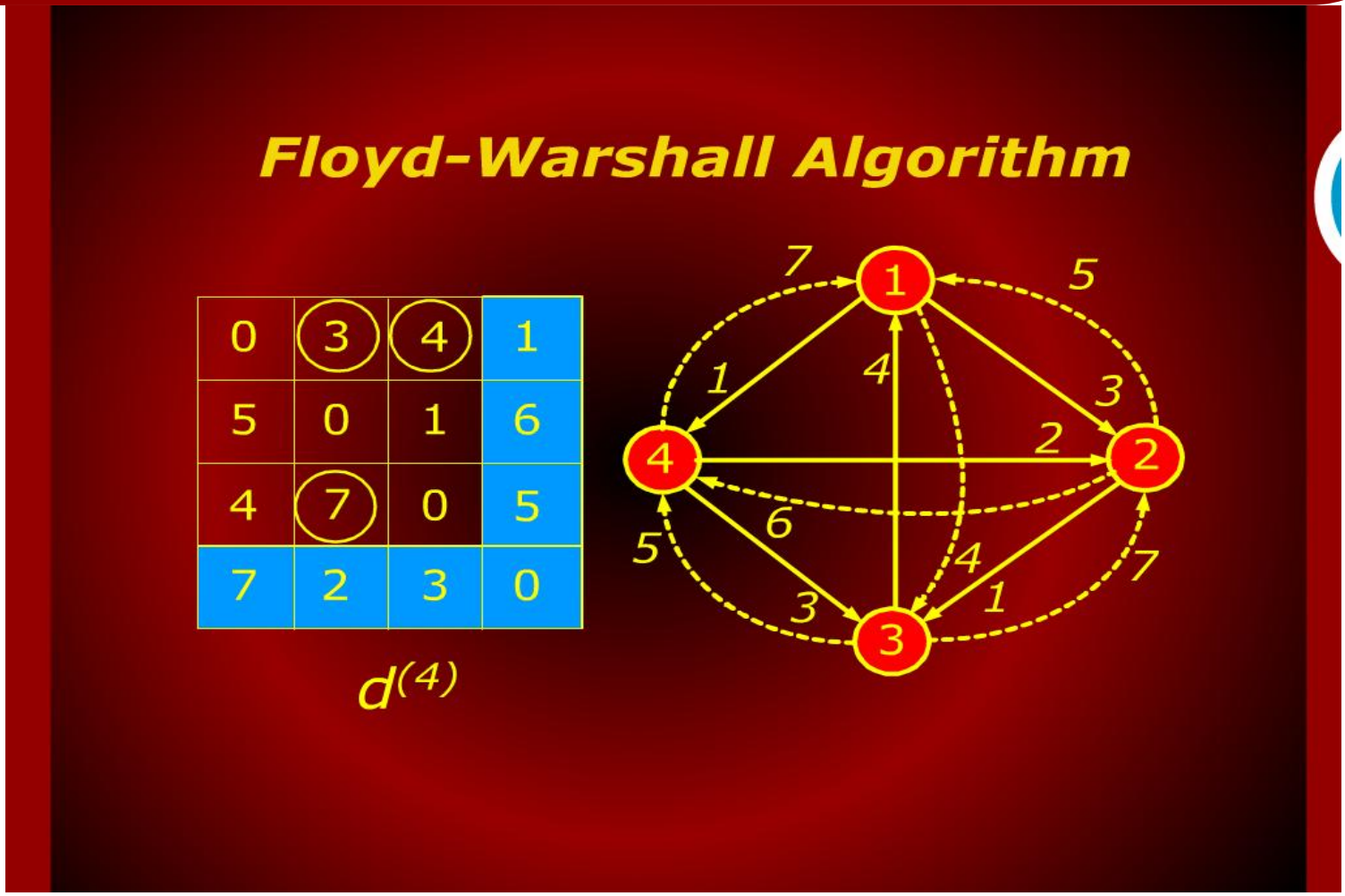

Criteria for analyzing Algorithms

Model of Computation